Fresh end-to-end checks of three ECDLP leads

This is a new execution, not a reconstruction of the unlocated historical runs. The complete claim-by-claim verdicts are in CLAIMS.md. Exact frozen observations are in results/, with source models, subgroup witnesses, input specs, sealed sessions and replay receipts.

The study covers 21 binary curves, 24 odd-characteristic coverage cases, and 18 PKM cells with 23 eligible primes each. Decomposition uses 3,520 target workloads. Genus-2 transfer uses two separate 1,260-target cohorts: one indexed by field size, and one by actual prime-subgroup size through 40 bits. Each genus target has four solver arms. Primary execution therefore comprises 23,952 solves, with 84 independent/unfiltered control solves and 168 supplementary trace-filter solves. Every completed answer is checked on its original elliptic curve; rejected fixtures and unsuccessful development attempts are preserved separately.

What the costs mean

All figures below are cold one-target operation-count observations. Algorithms receive P, Q and r, never the planted scalar. Charge destination construction, both transfers, base construction/projection, all initialization, roots and rational lifts, relation collection, elimination, failed work, retries and source verification. Common fixture generation/order certification is separately recorded.

Odd quadratic arithmetic expands into F_p M/S/I operations. Binary arithmetic uses the specified ambient-field M/S/I unit. Twelve arithmetic prices use S/M={0,0.5,1}, I/M={1,10,100,1000}. The retained ecbench GAE conversion uses the separately declared S/M=0.75, I/M=10. Native charges {0,0.1,1} source-addition equivalents apply symmetrically to both arms. These logical native counters do not measure allocations, physical hash buckets or CPU instructions. L0 walls are descriptive; no hardware-speed or fastest-tool claim follows.

The matched references are memoized r-adding rho with and without negation. Their collision-table memory is reported alongside the IC graph. `rho_neg` is the predeclared main reference; `rho_plain` remains visible throughout.

Genus-2 transfer

The implementation uses the published Scholten family and an explicit one-infinity quintic, $12Y^2=12+100t+299t^2+380t^3+187t^4+30t^5$. Pullback plus conjugation defines tau. The dual norm checks `norm(tau(P))=2P`; addition, exceptional points, descent and odd-prime subgroup preservation are checked. See TRANSFER.md, the exact polynomial certificate and small zeta/order checks. The certificate is an integer polynomial witness plus the repository's identity checker, not a Lean proof of the geometric theorem.

The primary collector discards repeated split divisor classes up to sign. It restarts on repeated walk divisors and cannot recover a scalar from those collisions. A signed negative cycle only anchors a large-prime component; extraction waits for a useful target dependence. Initial scalar work, restarts, all root work and graph/scalar operations remain charged.

Cohort / size label	Targets	IC/rho-neg, 12 arithmetic prices	Including native charge band	Declared default ratio	rho wins / 12
field / 16	180	3.811–7.044	3.613–7.044	5.465	12
field / 20	180	4.017–7.933	3.968–7.933	6.107	12
field / 24	180	4.396–10.422	4.396–10.422	8.127	12
field / 28	180	3.018–13.865	3.018–13.865	10.002	12
field / 32	180	2.879–28.203	2.879–28.203	17.806	12
field / 36	180	4.039–32.655	4.039–32.655	21.016	12
field / 40	180	4.600–39.658	4.600–39.658	25.054	12
subgroup / 16	180	3.398–7.779	3.398–7.779	5.990	12
subgroup / 20	180	2.409–9.295	2.409–9.295	6.611	12
subgroup / 24	180	1.928–12.555	1.928–12.555	8.521	12
subgroup / 28	180	1.596–15.293	1.596–15.293	9.692	12
subgroup / 32	180	1.466–14.861	1.466–14.861	9.249	12
subgroup / 36	180	1.496–17.300	1.496–17.300	10.603	12
subgroup / 40	180	1.421–18.362	1.421–18.362	11.091	12

The field cohort labels are approximate field-size levels; actual field and subgroup bits/cofactors are retained in the manifests. The order cohort requires exact subgroup bit lengths and cofactor at most 64. A 40-bit field is not a 40-bit prime-order subgroup. Tables report ratios of sums over matched targets, not a claim that every individual target has the same cost. Per-curve rows and all price cells remain in the machine-readable tables.

A generic full-degree Jacobian addition in the largest field fixture costs **70M+3S+I** in this backend; the matched source addition costs **9M+4S+I**, in base-field units. The supplied 41M+I Jacobian formula does not describe this backend; the source formula is reconciled below. Independent-trial and unfiltered controls, including all scalar-generation cost, are in genus-controls.json. First graph cycles versus useful solve positions are in genus-cycle-events.json.

Subfield and PKM decomposition

The binary linearized solver and odd S3 solver verify every rational lift by point addition. Factor-base logarithms are unknown variables; incremental elimination recovers the target. Cofactor offsets let relation collection visit the full relevant group, followed by projection into the prime subgroup. The corrected sampler tests initial/restarted targets and uses seeded counted jump choices; it does not assume a fixed point-partition map reaches a tiny sumset.

Family	Curves / targets	Per-curve IC/rho-neg across all 12 arithmetic prices	With native charge band	Whole-oracle M / decomposition call
Binary quadratic	21 / 168	2.089–70.311	2.089–113.842	12.141–16.187
Odd quadratic	5 / 40	2.079–3.505	2.079–3.505	46.270–150.778
Kummer/Dickson	414 / 3312	4.263–1695.104	4.263–1695.104	0.000–937.312

Binary q=512 mean full-group coverage is **0.373152**, compatible with approach to 3/8 over this range. Every x in each tested subfield has absolute trace zero; all 21 doubled images equal the trace kernel. This proves the finite coset obstruction, not a universal 3/8 limit. Prime-subgroup coverage is separately retained; a trace filter rejects no target in an odd-order subgroup. The 4.7% number compares two whole-group hit probabilities, not all relation multiplicities or the subgroup attack yield.

The 18 PKM cells have fresh maximum |z| **4.255** under the predeclared paired-prime statistic, with maximum absolute structured-minus-control coverage **0.00008987**. Thus the supplied |z|≤1.84 bound is not reproduced. Each curve has 32 sign-orbit/torsion-matched random controls. All effects and bootstrap intervals remain visible; failure to reject zero is not a proof of random equivalence.

The complete oracle census checks **9,378,482 targets across 854 curve/base combinations**, with zero disagreements against direct point-pair sumsets. Cost comparisons include complete solving on fixed tower fibers and matched random bases. This measures the implemented two-summand solving route; it does not exhaust every possible multivariate or auxiliary-isogeny PKM solver.

Reproduction and retained failures

Build the common engine with `cargo build --release --offline --bin ecbench`, and the standalone driver with `cargo build --release --offline --manifest-path research/three_leads_e2e_20261018/native/Cargo.toml`. The protocol and order-cohort protocol fix eligibility, seeds, caps and prices. `run-batch.sh` records each session; `verify --replay-all` re-executes every deterministic record. Archived sessions must be extracted to their original study-relative paths before replay.

The earlier v1 batches are diagnostic evidence, not silently discarded targets: unsplit-only genus walks and fixed-partition decomposition walks could exhaust. Their original implementations remain registered for exact replay. Corrected v2 cohorts use the same original models/targets/caps; the additional exact-order cohort is separately identified. The general-cubic coverage diagnostic and failed build/check attempts are retained as development evidence.

The report graph data are native Rust outputs. Python only renders the retained observations. All individual session reports and plots are indexed in the report bundle; the main plots and PDF are mirrored under `benchmarks/20261018-three-leads-e2e`. Canonical dashboard entries remain operation-model diagnostics outside fully priced speed rankings. No claim of fastest implementation or deployed-curve security is made.

Supplemental checks and implementation cost

The 168 additional binary trace-filter solves all verify and stop at exactly the same relation/trial positions as their unfiltered counterparts. The filter skips 3,044 of 6,046 candidate relations (50.35%). It tests relation-walk points in the full group; it rejects no odd-prime-subgroup DLP target. Its mask costs n(n−1) squarings once, followed by four logical bit operations per finite candidate. At q=512 it reduces the arithmetic-model cost to 0.971–0.987 of the unfiltered route; the native sensitivity band reaches 1.004. This is inexpensive filtering, not free setup. See trace-filter.json.

The source formula 9M+4S+I becomes exactly 12M+I at S/M=0.75. Thus the supplied source-addition formula is compatible with that convention; 41M+I does not describe this Jacobian backend.

In the PKM decomposition pipeline, 562 of 6,624 structured/random-base returns occurred before any decomposition, including 64 empty-base returns. Those are generic identity hits rather than relation-based IC successes. All costs include that behavior; the result does not certify a pure IC mechanism on every tiny target. See decomposition-mechanisms.json.

The supplementary uniform-random-x null (32 controls per curve, added after the conditional analysis) gives maximum absolute PKM z=2.867, compared with 4.255 for controls conditioned on actual point count. Neither reproduces 1.84. These nominal cell standardizations are not a multiplicity-adjusted universal significance claim.

At actual 40-bit subgroup size, complete instrumented solves retire 7.4868 times as many user-space instructions as negation rho (paired curve/target bootstrap 95% interval 6.6571–8.3716), or 5.7823 times plain rho (5.2151–6.4286). All 180 pairs use the same binary and valid unmultiplexed counters. These counts include native work and instrumentation overhead. A five-pair identical-rho A/A control has ratio range 0.9999703–1.0000121 and identical primitive/native counters. See retired-instruction-summary.json and instruction-AA.json. Both attempted L2 timing preflights rejected a busy host; no qualified wall-time result is claimed.

All 945 sealed study sessions pass their audits. Every one of 42,862 successful measured records replays identically. The 134 exhausted development records were separately rerun and reproduce their exhaustion and exact counters; the standard replay command skips unsuccessful records. The A/A session adds ten measured records and two warmups separately.

The complete evidence and reproduction guide documents the deterministic raw archive, exact replay commands, source snapshots and validation limits. Actual field/subgroup bit lengths and cofactors for all 42 genus models are in genus-fixture-sizes.csv.

Retired instructions favor the tiny subfield pipeline in some cases despite its arithmetic-model loss: binary q=8 gives 0.4491× rho-neg, q=16 gives 0.4942×; odd p=5 gives 0.4363× and p=7 gives 0.4854×. All 18 structured PKM cell instruction ratios exceed one (1.928–8.190). These complete instrumented-backend observations are a separate unit; they establish no qualified wall-speed or fastest-implementation claim.

Recorded observations

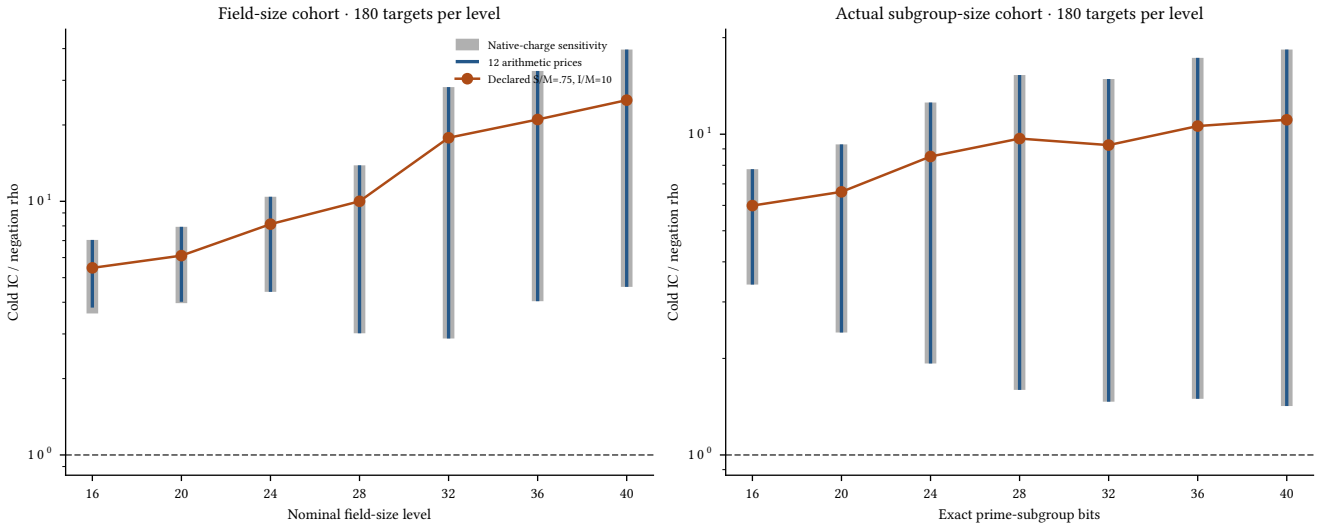


Figure 1: Price sensitivity, not a confidence interval. Cold work includes setup, transfer, roots, elimination and source verification. Each point uses 180 matched targets; both cohorts contain 1,260 targets.

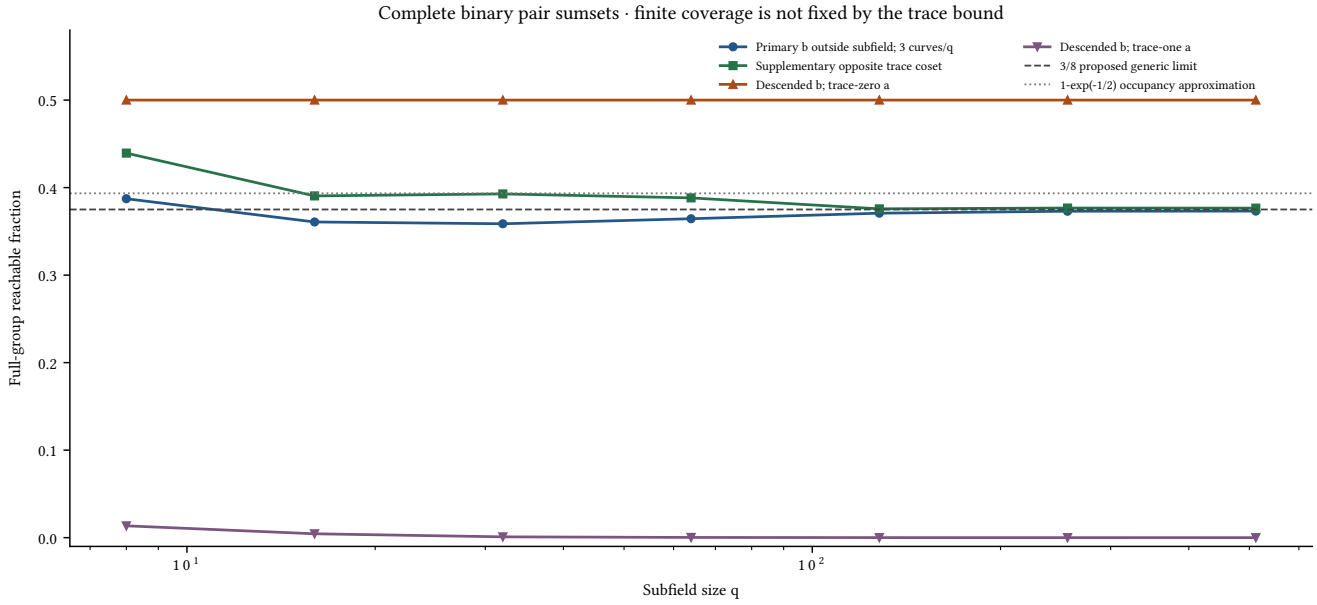


Figure 2: Primary and supplementary complete finite sumsets. Descended-b controls demonstrate why the trace kernel alone does not establish a 3/8 limit.

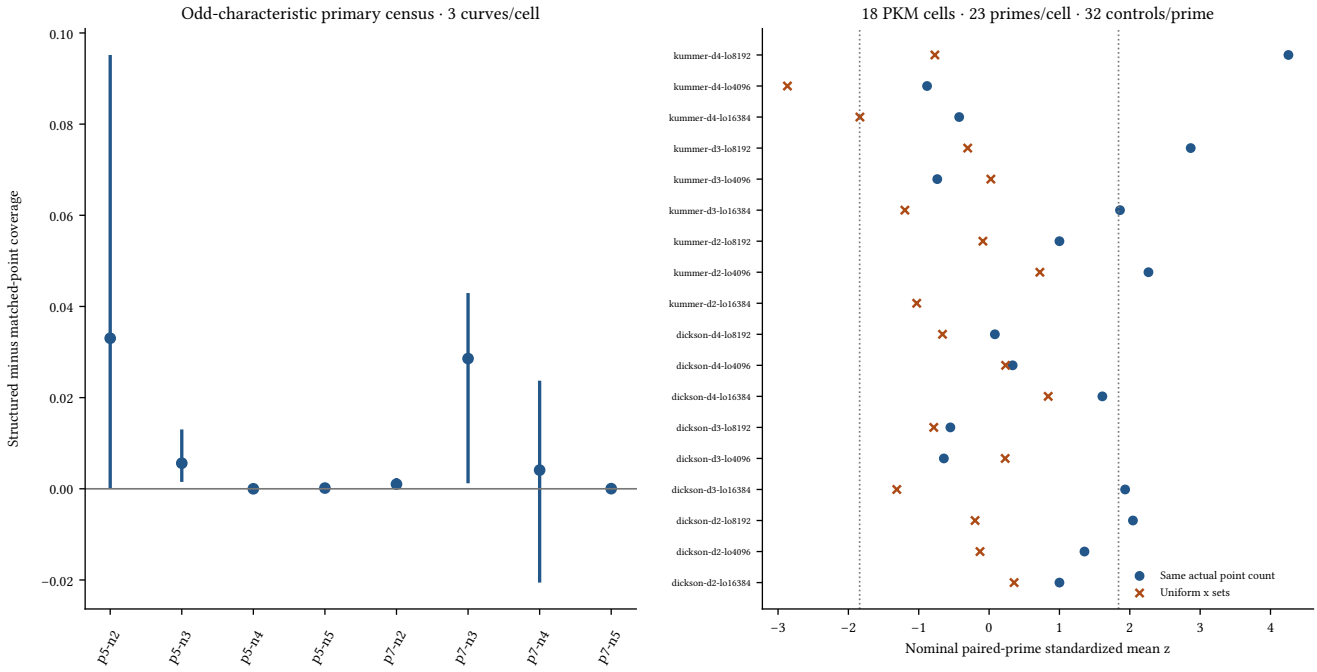


Figure 3: The two PKM nulls answer different questions. Odd-cell bars are native curve-resampling intervals. Standardizations and intervals do not prove universal random equivalence.

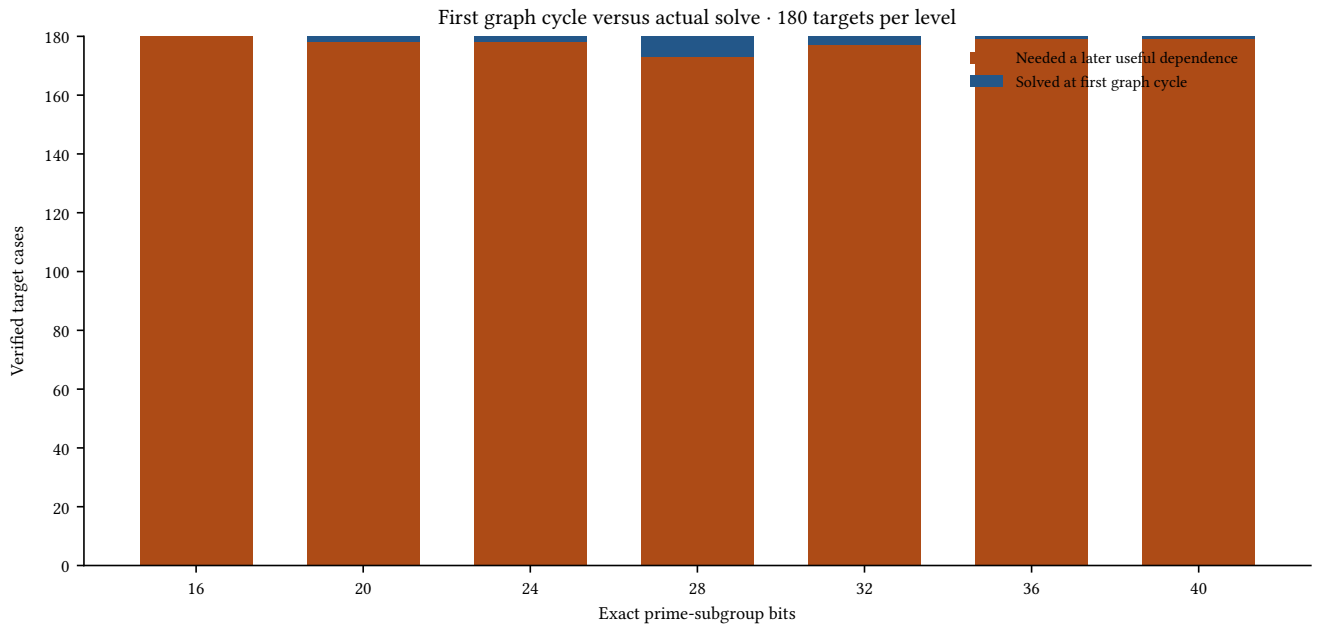


Figure 4: The signed collector distinguishes graph cycles from useful target dependencies. Repeated split divisor classes are discarded in the primary arm.

Verdict on every original claim

These verdicts concern the supplied claims and the fresh implementations in this PR. The historical producer, seeds, pricing definitions and raw 1,260 runs remain unlocated. Fresh agreement does not authenticate an exact historical statistic. All cost intervals below are ratios of sums over matched targets at explicitly declared prices, not confidence intervals. Scope is the tested synthetic families and native backends.

1. Subfield decomposition

Original claim	Verdict	Fresh evidence and qualification
Odd characteristic n=2 through 5 matches a random base of the same size.	Broadly compatible in small tested cases; equivalence not established.	24 primary curves: p=5,7; n=2,3,4,5; three curves/cell; 32 actual-point-count matched controls/curve. Maximum nominal absolute z=2.087. Seven of eight cell intervals lie within a loose ±0.05 coverage margin; the remaining p=5,n=2 interval extends to about +0.095. Random-x and matched-point nulls are separately retained.
Weil descent creates no extra relations there.	Not proved by coverage.	A finite coverage census tests reachable targets, not relation multiplicities or every solver. Eight descended-coefficient odd controls demonstrate that model selection matters. No universal theorem follows.
Binary n=2 is the exception.	Too narrow as a theoretical statement.	Trace from an even-degree extension annihilates every subfield element. The same character obstruction applies to fixed even summand counts in other even extension degrees under the normalized ordinary binary model. Only n=2 is costed end to end here.
Binary coverage tends exactly to 3/8.	Finite trend reproduced; universal limit unproved.	Mean full-group coverage at q=512 is 0.3731522954 on three primary non-descended curves; opposite-coset controls give 0.3765374531. Descended-b controls approach 1/2 or give only the identity (1/N), so the trace restriction alone cannot imply 3/8. Seven q levels, three primary curves each.
Expected random coverage is 0.3935.	Occupancy approximation, not an exact finite null.	1−exp(−1/2)=0.3934693403 is the usual heuristic at this density. Actual matched random-base mean at q=512 is 0.3920743848. Full finite counts and 32 controls/curve replace the approximation in comparisons.
z=−4.7.	Exact historical statistic unverified; not reproduced.	Fresh q=512 paired-curve nominal z=−160.3900 under the declared control construction. Only three deterministic primary curves at that level; this is not the original statistic or a calibrated universal significance statement.
Every x in F_q has trace zero over $F_{(q^2)}$.	Verified, with trace convention explicit.	Absolute trace to F_2 is zero. Relative trace to F_q is also zero in characteristic two. The group character is absolute Tr(x+a), not simply an unqualified trace of x.
The whole factor base sits in one coset of 2E; checked on 21 curves.	Verified for the ordinary normalized binary model.	On all 21 primary curves, exhaustive doubled images equal the character kernel and every rational lift of subfield x has one fixed character. Two summands therefore land in 2E, whatever that coset is. See the retained trace checks and complete census.
Only half the group can ever decompose.	Correct upper bound for the full group, not exact coverage.	Two-summand targets belong to the index-two subgroup 2E. The odd-prime DLP subgroup already lies inside 2E, so half of the DLP targets are not ruled out.
About 4.7% fewer relations per target.	Misstated unit.	(0.3934693403−0.375)/0.3934693403 is about 4.69%, a comparison of whole-group hit probabilities. It is not a measured reduction in relation multiplicity or subgroup DLP yield. Fresh measured whole-group coverage is retained without that substitution.
A free trace test skips impossible targets.	Useful filter; setup is not free.	168 additional end-to-end solves retain identical stopping positions. The test skips 3,044/6,046 full-group relation candidates; none of the odd-order DLP targets fails it. A cold n(n−1)-squaring mask and logical per-candidate work are charged. At q=512 arithmetic cost is 0.971–0.987 of the unfiltered route; the native sensitivity reaches 1.004.
End-to-end n=2 costs 2.7–66× rho in every pricing cell.	Arithmetic loss reproduced; numerical range replaced.	168 binary and 40 eligible odd-quadratic targets, all verified. Per-curve binary pipeline/negation-rho ratios span 2.089–70.311 over the 12 arithmetic prices (up to 113.842 with native sensitivity). Odd quadratic spans 2.079–3.505. Original prices/reference were unavailable. Every tested per-curve arithmetic cell loses to rho. Tiny binary q=8,16 and odd p=5,7 pipelines use fewer instrumented retired instructions than rho-neg (0.4491, 0.4942, 0.4363, 0.4854 respectively); that is a separate unit, not a qualified wall-speed claim.
Root finding costs about 1,000 multiplications per call.	Not an intrinsic average requirement.	Complete oracle M/call averages range 12.141–16.187 in binary and 46.270–150.778 in odd quadratic. These include rational-lift checks but list squarings/inversions separately. They are not full M-equivalent prices or worst-case root bounds. Binary linearized Artin–Schreier solving avoids a generic degree-polynomial root routine.

2. Prime fields and PKM tower bases

Original claim	Verdict	Fresh evidence and qualification
Kummer and Dickson: 18 cells, at least 23 primes each.	Fresh design executed.	Two families × depths 2,3,4 × three prime bands; exactly 23 eligible primes/cell, 414 fixtures. Rejections, construction specs and exact fiber bases retained. This does not recover the historical 18 cells.
Coverage indistinguishable from random x sets; absolute z≤1.84 everywhere.	Bound not reproduced; effects are small.	Maximum absolute nominal z=4.2548 under controls matched by actual point count, maximum absolute coverage difference 0.0000898746. Supplementary unconditional uniform x sets give maximum absolute z=2.8665. Nulls differ; the latter was added after the conditional analysis. Neither bound is 1.84; no unadjusted cell statistic establishes a universal effect.
Tower structure creates no extra relations.	Universal statement unsupported.	Coverage differences do not count all relations or prove random equivalence. Complete fixed-fiber and random-base sumsets are available. Small departures occur; no useful end-to-end advantage appears for this implemented two-summand route.

Original claim	Verdict	Fresh evidence and qualification
Any PKM advantage must come from solving, which was not measured.	Fresh solving measured; other solvers remain open.	3,312 targets, both structured and matched-random pipelines plus plain/negation rho; 13,248 solves, all verified. Structured pipeline arithmetic ratios span 4.263–1695.104 across per-curve cells versus negation rho. Of 6,624 pipeline returns, 562 precede any decomposition, including 64 empty-base returns: generic identity hits, not pure IC evidence. This tests fixed Kummer/Dickson fibers, not every auxiliary-elliptic-isogeny or multivariate solver from PKM.

3. Genus-2 transfer and index calculus

Original claim	Verdict	Fresh evidence and qualification
C/F_p has Jacobian isogenous to the Weil restriction of E/F_(p^2), with a checked transfer tau.	Implemented for an explicit published family.	Scholten/Bernstein–Lange construction; one-infinity quintic $12Y^2=12+100t+299t^2+380t^3+187t^4+30t^5$. Pullback plus conjugation and the dual norm satisfy norm(tau(P))=2P. Exact integer polynomial cover certificate, exhaustive small-field map/norm checks and independent zeta/order counts on eight fields; 32 seeded map/additivity checks on each of 42 primary models. The geometric theorem is sourced, not newly formally proved.
1,260 runs through 40 bits, each checked by k·P.	Fresh requirement exceeded, with size ambiguity resolved.	Two separate 1,260-target cohorts (field-indexed and exact prime-subgroup-indexed), four arms each: 10,080 primary genus solves. All pass source-curve verification. Actual subgroup sizes are exactly 16,20,24,28,32,36,40 bits in the second cohort; its largest fields are 44 bits. The first cohort's nominal field labels must not be read as subgroup bits.
IC solves through the first large-prime cycle.	False for the implemented signed graph.	In the actual-order cohort, 1,244/1,260 solves require a later useful dependence; only 16 solve at the first graph cycle. Negative cycles can anchor a large-prime variable without identifying k. At actual 40 bits, 179/180 solve later.
IC uses 0.27–0.37·p Jacobian operations.	Search-count trend reproduced; cold count differs at small sizes.	Actual-order mean walk trials/p range 0.27479–0.31520. Cold Jacobian operations/p range 0.92912 at 16 bits down to 0.31268 at 40 bits. Setup, scalar initialization, transfer and verification cannot be dropped when calling this end to end.
Rho needs 0.31–0.44·p operations.	Mostly consistent with plain-rho search at larger tested sizes; reference matters.	Plain-rho walk trials/p are 0.32092–0.33669 from 20 through 40 bits; 16-bit value is 0.25287. Actual 40-bit cold source operations/p=0.32406. Stronger negation-rho search is 0.23709·p there. Counts scale with sqrt(r), so p-normalization also depends on the cofactor.
Independent random trials give the same count.	Compatible counts, not demonstrated equality.	21 independent controls per cohort: order normalized trial ratio 1.04439 (curve-bootstrap 95% 0.89882–1.20455); field ratio 0.89860 [0.73878,1.09754]. Order-cohort independent trials cost 25.766× primary at default prices, including fresh scalar generation. Another 21 unfiltered controls per cohort are separate.
Therefore this is not rho hiding in the walk.	Mechanism audited; statistical argument alone insufficient.	Primary extraction uses signed relation dependencies; repeated walk states trigger charged restarts rather than rho extraction, and repeated split divisor classes are discarded up to sign. Independent sampling is a separate charged control. Equal-looking stopping counts alone would not establish the mechanism.
A Jacobian addition costs 41M+I.	Not this backend.	100 generic full-degree additions on the largest fixture each cost 70M+3S+I in base-F_p units. Cantor exceptional cases use a checked fallback. This is an implementation measurement, not a lower bound on optimized Jacobian arithmetic.
E(F_(p^2)) addition costs 12M+I.	Compatible at a stated squaring price.	Generic source additions cost 9M+4S+I in base-F_p units. At S/M=0.75 this is exactly 12M+I; other price cells differ.
Ranking unresolved at every size; cost ratio 0.51–2.83.	Not reproduced; resolved within the declared fresh model.	All 14 cohort/size means lose to negation rho in all 12 arithmetic cells. Actual 40-bit subgroup IC/rho-neg=1.42081–18.36249; IC/plain-rho=1.07626–13.93230. Native sensitivity preserves the actual-40-bit range. This is a finite implementation/model verdict, not a proof that every genus-2 algorithm loses.
At 40 bits rho wins 10 of 12 cells.	Replaced by 12/12.	Both source references win all 12 arithmetic cells at actual 40-bit subgroup size; negation rho also wins all 12 at nominal field-40. Ratios use complete cold work.
Measured gap about 2.8×.	Not reproduced in these observations.	Actual-order-40 default arithmetic gap=11.09091× versus negation rho, 8.40978× versus plain rho. Complete instrumented retired-instruction gaps=7.48684× [6.65709,8.37164] and 5.78231× [5.21506,6.42857], respectively. Five-pair rho A/A range 0.9999703–1.0000121. L2 timing preflights rejected host contention; no qualified wall-time gap is asserted.

The original operation formulas also impose a consistency check. If both quoted operation-count ranges are complete and use a common nonnegative I/M price, their simplified IC/rho ratio is at least (0.27/0.44)·(41+I)/(12+I), hence at least 27/44 (about 0.614). A value of 0.51 therefore needs different populations, extra reference costs or a different accounting convention; it cannot be derived from those four supplied ranges alone.

Scope of completion

All three requested routes have fresh source-to-answer executions, not coverage-only proxies. The complete rational-decomposition oracle census has 9,378,482 targets across 854 curve/base combinations and zero disagreements. All 23,952 primary solves, 84 independent/unfiltered controls and 168 trace-filter controls verify. Development failures remain sealed: all 42,862 successful measured records replay identically and all 134 exhausted records independently reproduce their exact counters and failure state. The additional A/A session is separately indexed.

Neither a finite experiment nor this implementation exhausts all mathematical algorithms. Exact historical numerical authentication remains unavailable without the original producer and raw data. The fresh numbers above resolve what this PR actually measures and make every supplied assertion reviewable.